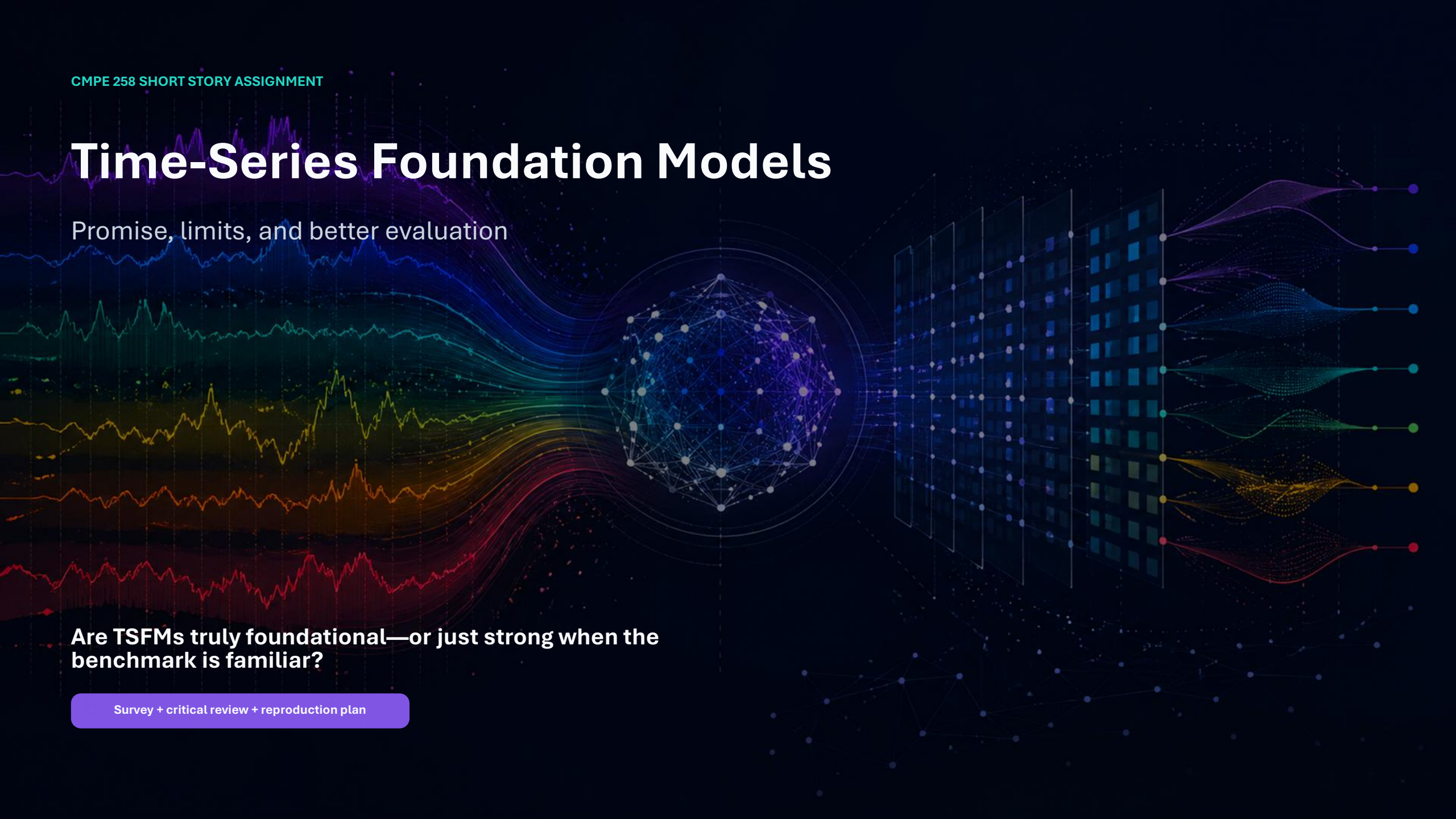


Time-Series Foundation Models

Promise, limits, and better evaluation

Are TSFMs truly foundational—or just strong when the benchmark is familiar?

Survey + critical review + reproduction plan



Three beats of the argument

The story is not just bigger models. It is promise, evidence, and evaluation hygiene.

1. Promise

Pretrain once, adapt across many forecasting domains.

2. Reality check

Zero-shot strength depends on pretraining-domain match.

3. Evaluation

Leakage can make benchmarks look cleaner than they are.

Takeaway: TSFMs are promising, not proven universal.

Time series is not one world

A stock chart, ECG trace, weather series, and sensor stream are all sequences—but they come from different systems.

- Different scales and units
- Different noise and seasonality
- Different causal mechanisms
- Different leakage risks

That makes “one model for everything” much harder than it sounds.



What is a TSFM?

A pretrained sequence model intended to transfer across forecasting tasks, datasets, and domains.



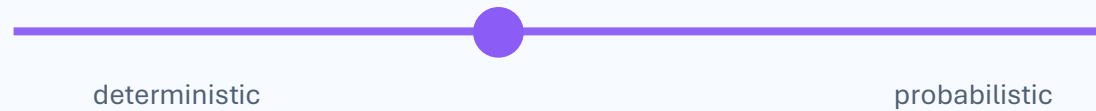
The claim: temporal patterns learned at scale can become reusable.

Survey taxonomy: how TSFMs differ

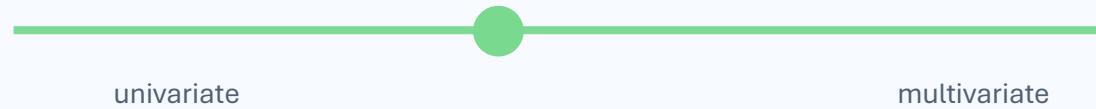
Input representation



Prediction style



Data handling



Training objective



The field is already diverse. “TSFM” is a category, not a single architecture.

The promise is real—but conditional

The recent papers turn the story from “one model forecasts all” into a more careful question.

What people hope

Zero-shot transfer everywhere

Large scale always wins

Benchmarks prove generalization

Pretrain once, reuse anywhere



What the papers suggest

Transfer depends on domain match

Smaller tuned models can compete

Leakage can inflate benchmark scores

Reusability needs clean evidence

Thesis: the word “foundational” is an aspiration, not yet a settled fact.

Reality check: domain alignment drives transfer

- Zero-shot strength is tied to pretraining domains
- Unseen domains can weaken the “foundation” claim
- Model size and memory cost matter in deployment
- Smaller specialized models can remain competitive



Question to ask: Did the model generalize—or was the target domain familiar?

Transfer depends on overlap between worlds

The practical value of TSFMs rises when the downstream domain resembles pretraining—and falls when it does not.

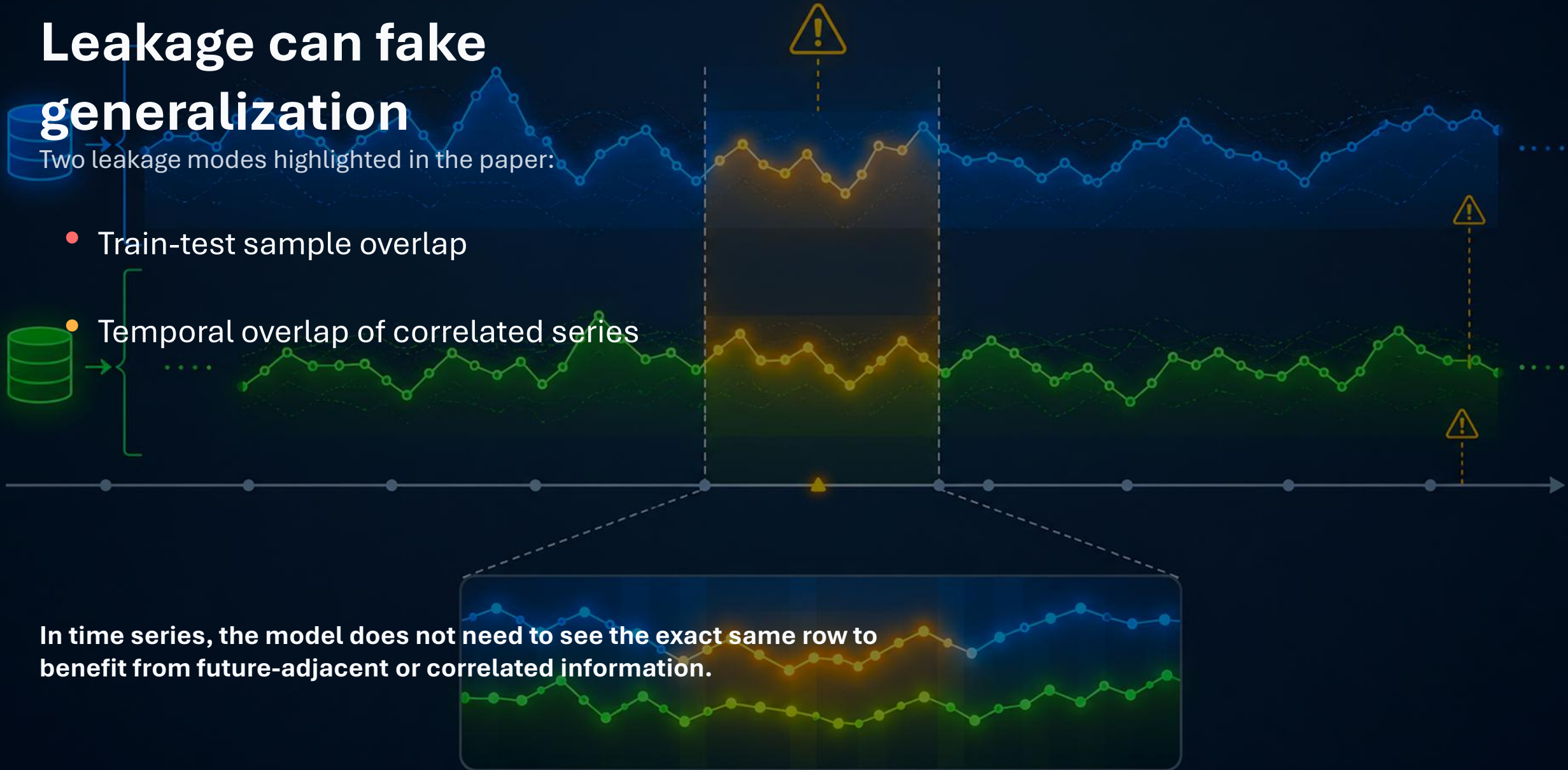


The farther the downstream task is from pretraining, the weaker the transfer claim becomes.

Leakage can fake generalization

Two leakage modes highlighted in the paper:

- Train-test sample overlap
- Temporal overlap of correlated series

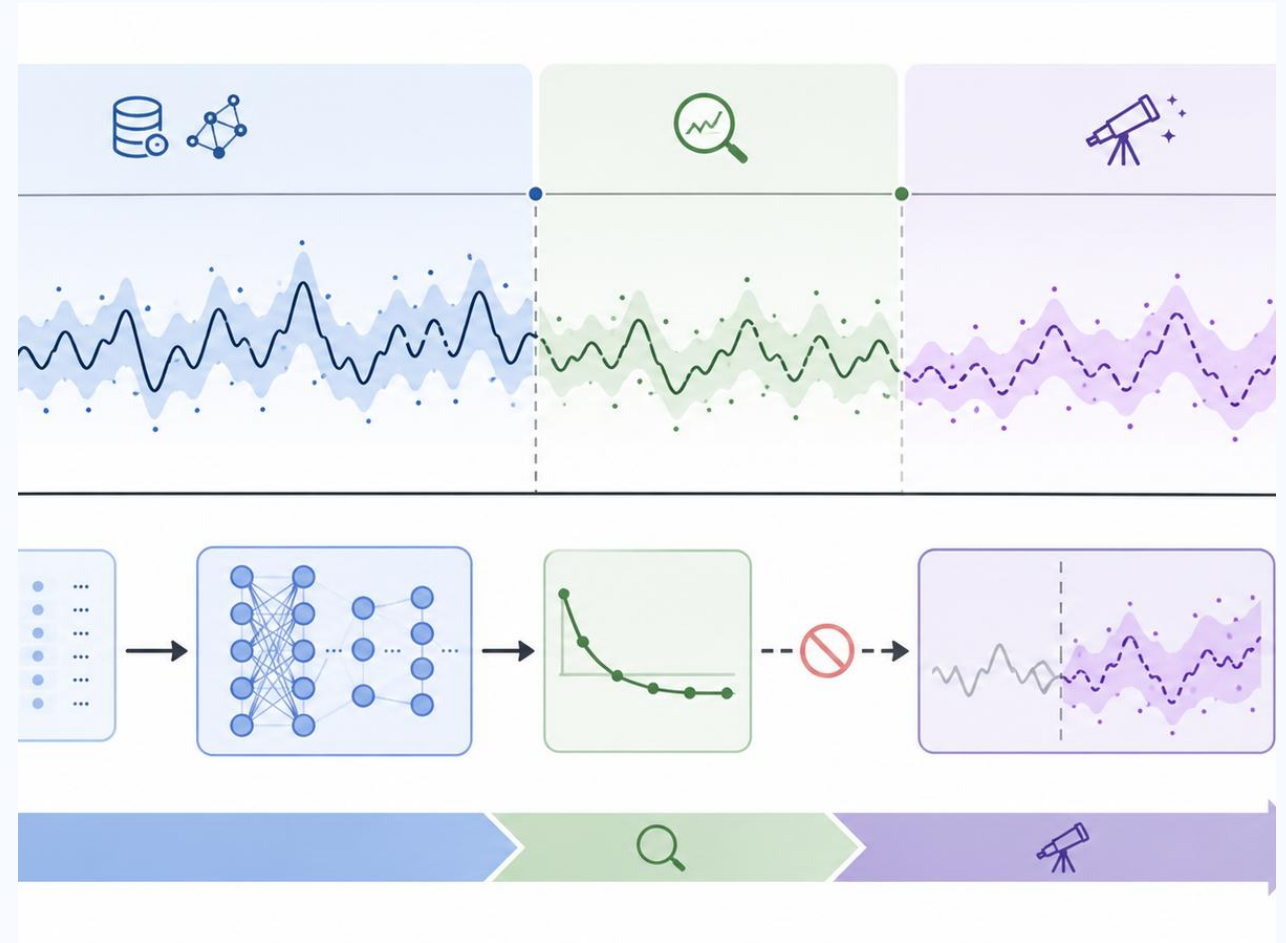


In time series, the model does not need to see the exact same row to benefit from future-adjacent or correlated information.

What better evaluation looks like

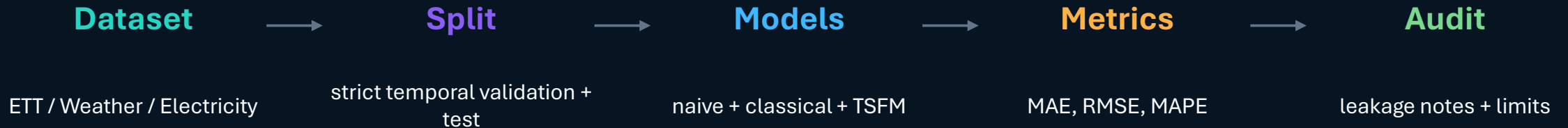
- Respect chronology
- Separate pretraining and benchmark data
- Audit correlated overlap
- Compare against strong smaller baselines
- Report compute and memory tradeoffs

Benchmark design is part of the scientific claim.



AutoResearch-style experiment

A small reproduction focuses on the paper's core message: do gains survive clean evaluation?



What to report in the repo

- configuration + runs
- metric tables and plots
- split details and leakage checks

Fill this after running your experiments

Model	MAE	RMSE	MAPE	Observation
Naive / seasonal naive	TBD	TBD	TBD	TBD
Classical baseline	TBD	TBD	TBD	TBD
Small neural model	TBD	TBD	TBD	TBD
TSFM zero-shot	TBD	TBD	TBD	TBD
TSFM fine-tuned	TBD	TBD	TBD	TBD

Interpretation prompts

Did TSFM gains survive strict temporal splitting? Did a smaller baseline compete? Were results domain-dependent?

How to use TSFMs responsibly

The safest interpretation is not “ignore TSFMs.” It is “evaluate them like the benchmark can lie.”

Trust, but audit

Check pretraining data and possible overlap.

Match the domain

Transfer is strongest when worlds are similar.

Use strong baselines

Small tuned models are not automatic strawmen.

Report tradeoffs

Accuracy needs context: compute, memory, robustness.

The best TSFM result is not the highest number—it is the highest number that survives honest evaluation.

TSFMs are promising, not proven universal.

Their real value depends on domain alignment, clean evaluation, and whether gains justify model cost.

Main references

Foundation Models for Time Series: A Survey — [arXiv:2504.04011](#)

How Foundational are Foundation Models for Time Series Forecasting? — [arXiv:2510.00742](#)

Rethinking Evaluation in the Era of Time Series Foundation Models — [arXiv:2510.13654](#)

Thank you